

Sound to Sight: Enhancing Contextual Awareness for the DHH Community in Public Transportation

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Abstract

Public transportation systems remain significantly inaccessible to Deaf and Hard of Hearing (DHH) individuals, who often miss critical auditory information like announcements and alarms, impacting their safety and autonomy. Current assistive devices and mobile applications fall short in providing real-time alerts tailored to their needs. This paper presents a system integrated with Google Maps to enhance auditory awareness for DHH users using sound recognition and multimodal notifications. Grounded in user-centered research, including interviews with DHH individuals and accessibility experts, the design introduces features like animated sound alerts, audio history logs, and user-validated sound feedback – these prioritize accessibility, usability, and inclusivity. Expert evaluations confirmed the prototype’s potential to improve navigation and foster independence for DHH individuals. This work highlights the importance of context-aware design in creating equitable urban spaces and sets the foundation for broader applications beyond public transportation.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; **HCI design and evaluation methods**.

Keywords

Deaf and Hard of Hearing, Contextual Awareness, Google Maps, Auditory Cues, Sound Alerts

ACM Reference Format:

Rachit Bhayana, Timothy Chiu, Natalie Jarrett, Disha Sikaria, and Lynn Dombrowski. 2025. Sound to Sight: Enhancing Contextual Awareness for the DHH Community in Public Transportation. In *Extended Abstracts of*

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720305>

the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 5 pages.
<https://doi.org/10.1145/3706599.3720305>

1 Introduction

Urban accessibility efforts increasingly emphasize inclusivity, yet public infrastructure remains inaccessible for DHH individuals, particularly in public transportation [1]. DHH individuals rely on visual cues, but essential environmental signals like announcements and sirens are transmitted auditorily, affecting their safety and autonomy [5].

Although many DHH individuals regularly use assistive technologies, existing solutions may fall short in addressing auditory needs in public spaces [13]. Hearing aids, a popular intervention among DHH individuals, are not always accurate in identifying sounds occurring outside their field of sensing, and as a result, the individual must take great caution in activities such as crossing roads due to their “inability to perceive noises of approaching vehicles” [20]. Crossing the road is an outdoors activity, where noise is understandably loud and crowded, but this issue applies to both outdoor and indoor spaces. Additionally, poor acoustic design can lead to “listener’s fatigue” [20] since DHH individuals must put in serious cognitive effort to distinguish between background noise, speech, and dangerous noises. This constant effort can lead to mental drain, stress, and reduced social participation.

Through our primary research with DHH community members, educators, and accessibility members, we found that navigating public transportation can be a particularly frustrating experience. DHH individuals may struggle to receive critical information such as speaker announcements or may have safety concerns due to inability to hear incoming vehicles or safety alarms. Many of our research participants resided in urban environments (i.e. Atlanta, New York City) where traffic density may incentivize residents to seek alternative modes of transportation, thus it is critical to ensure that public transport is accessible for people of all abilities.

Our system seeks to improve the experiences of DHH individuals using public transportation by providing real-time alerts for critical background sounds. We found that while many DHH individuals are open to new technology, they may be frustrated by inconsistent accessibility features [14] and by having to shift between multiple

different tools and applications. To ease the burden of technology overload, we integrated our system into Google Maps, a familiar and widely used application.

Overall, we approached our project committed to the idea that public transportation should be accessible to all regardless of hearing ability. We believe that by providing emergency notifications, automated announcements, and incoming vehicle alerts are critical we may help bridge the accessibility gap in the use of public transportation.

2 Related Work

Design for disability has evolved through modern technologies. Historically, few laws ensured care for people with disabilities, prompting advocacy focused on independence. However, solely prioritizing independence limits innovation that addresses individual needs [3].

Independence in disability design empowers decision-making through supportive infrastructure [3]. However, Bennett et al. propose focusing on interdependence, emphasizing relational interactions. In a study on AI-enabled assistive technology, interdependence was crucial for successful navigation involving partnerships between individuals with disabilities and guides [21]. Considering both independence and interdependence fosters more accessible, equitable designs.

Sound awareness for individuals with hearing impairments predominantly relies on visual and tactile cues, which may result in the omission of crucial auditory information. Early platforms enhancing auditory awareness used stationary systems to visualize sound characteristics but lacked categorization capabilities. Ho-Ching et al. used designs to analyze pitch, amplitude, and spatial positioning of sounds [7].

Matthews et al. proposed design criteria for non-speech audio cues, focusing on ease of interpretation and sound identification [17]. Bragg et al. developed a mobile app that alerts users to relevant sounds through haptic and visual feedback but faced limitations in sensor capabilities and interface design [4]. Matthews et al. created a digital app providing transcriptions of environmental sounds from the last 30 seconds [16].

Jain et al. developed an app for personalized sound recognition using user-provided audio clips [10]. ProtoSound adapted across contexts, delivering real-time results. They also created a smartwatch app using deep learning to detect sounds, optimizing latency with a watch-phone pairing but facing challenges in accuracy and user-defined categories [12]. Huang et al. evaluated a smartwatch app for recognizing background sounds, identifying challenges in noisy environments [8].

Goodman et al. explored visual and haptic feedback for contextual sound awareness. Participants preferred limited haptic feedback to manage complexity [6]. Mielke and Bruck developed a smartwatch detecting ambient sounds and providing alerts, emphasizing added functionalities for performance [18]. Bandara and Balasuriya designed a roadside threat alert system using microcontrollers and vibrating motors, demonstrating high accuracy at low cost [2].

Researchers have explored methods to augment background audio cues. Marino et al. used particle animations in video conferencing platforms to represent sound characteristics [15]. Jain et al. created a personalized sound awareness system for smart homes, addressing privacy and cognitive overload concerns [11]. Huang et al. introduced an AI-driven auditory awareness system to help DHH individuals correlate sound properties with context, improving sound recognition [9].

3 User Research

To understand the challenges faced by DHH individuals in navigating public transportation and to design tailored accessibility solutions, we conducted user research that combined secondary research with direct user engagement. Building on the insights gained through a review of scholarly literature and accessibility reports, we conducted six semi-structured interviews and one unstructured interview, each lasting approximately one hour, with individuals representing diverse perspectives within the DHH community. Our participants included three individuals identifying themselves as Deaf, an ASL teacher, an executive director of a Deaf advocacy organization, a board member of a Deaf advocacy organization, and one Hard of Hearing individual. These conversations offered personal insights into daily accessibility challenges, systemic barriers, and unmet needs. To enrich our understanding, we also interviewed two researchers specializing in DHH studies and one accessibility designer, who provided perspectives on addressing accessibility gaps and designing inclusive systems.

We found several important points from the interviews relating directly to our system. Challenges arise from excessive ambient noise within large occupancy public spaces, which makes communication difficult. Additionally, barriers to communication are common due to the fear of being misinterpreted. Many participants rely upon visual communication tools such as Zoom, Video Relay Services, and FaceTime. The user research revealed the critical need for real-time visual alerts to supplement auditory announcements, the importance of multimodal navigation aids that accommodate diverse user preferences, and the value of collaborating with advocacy groups to address systemic issues. These findings informed our design goals and shaped the development of inclusive solutions that prioritize the safety and autonomy of DHH individuals.

4 Design Requirements

The requirements we defined after concluding our research findings as depicted in Table 1.

5 Concepting & Ideating

For our concept refinement, we conducted a co-design session with an accessibility researcher.

Our team divided roles for the co-design: two facilitators and two notetakers. The facilitators introduced the problem and research conducted to the participant. They then listened to the participant's ideas for solutions, sketching out ideas alongside them. The participant was encouraged to think aloud and to share ideas in real-time. Meanwhile, notetakers jotted ideas and independently sketched additional ideas based on their thoughts and notes in the process.

Table 1: Design Requirements and Rationale

Requirement	Rationale
Convenience The design should avoid intense and repetitive notifications. The design should not require extra technology or devices.	Users mentioned that many notifications could be difficult to manage and end up being ignored. They emphasized not wanting to carry extra devices, citing the inconvenience of devices such as hearing aids and augmented reality glasses.
Intuitive and Streamlined Our design should be friendly for non “tech-savvy users” and should include customizability options.	Given that some DHH individuals classify as older adults, there should be clear onboarding, and the design should be intuitive. Customizable designs allow users to change notifications to what they want to be alerted about, which is especially important in our design, to allow users to prioritize different safety needs.
Delight The design should include moments of delight throughout the experience.	Interviewees expressed interest in concepts that were optimistic. They want to be safe, but the application should not create stress or fear for them.
Accessibility Our design should support multi-modal cues. Our design must utilize interdependent approaches to accessibility design.	Participants shared that if visual notifications were accompanied with haptics, they could react quicker to their situation. Participants expressed discomfort with relying entirely on AI software and would like human validation.

Insights shared emphasized ethical issues with our design, such as the possibility of automatic activation of the audio-feature, which could feel like a privacy violation. The user should have control over when the tool would be “listening”. To be inclusive of individuals with varying levels of literacy, we were recommended to avoid reliance on text. Instead, visual symbols and illustrative alerts could be deployed to convey different levels of urgency to be more accessible for people with different abilities. We conceptualized five to seven visual cues in this session which include automated announcements, approaching public transportation vehicles, fire alarms, change of platform or track, and other important cues observed on a transportation journey. Such cues would be represented with distinct illustrations such as a “train approaching” or “bus nearby.” Transcriptions of announcements and specific illustrations for alarms and other relevant elements were also recommended to enhance awareness for DHH users.

The co-design session gave us a thoughtful and inclusive framework to begin designing an accessible tool that visualizes audio cues, ensuring that the design emphasizes safety and usability, catered to the unique needs of DHH users.

Next, we sketched ideas according to the features and user requirements outlined from the co-design and previous research methods conducted. Afterwards, we selected the most effective and relevant elements to be integrated into our final design.

While we had formalized our concept, further evaluation was required to ensure that the design meets the expectations of DHH users. Our feedback session with Deaf-Identifying participants revealed that the animated visual alerts were a pleasant addition as they helped capture attention to the notifications received. They appreciated the visual design of the notification card and its announcement transcription feature. The participant pointed out that an issue with the design was that it relied on color to define noise levels, which is inaccessible to color-blind users. Additionally, reliance on the AI transcription model for audio cues was concerning as accuracy in detection is not necessarily guaranteed. The participant suggested the visual cue for an ambulance might be a welcome addition given emergency situations.

6 Design Details

6.1 Visual Design Language

Since we chose to nest our product in Google Maps as a feature, our visual design language included many existing assets of Maps, and any additional elements that we created that were inspired by Maps and other Google products, such as Live Transcribe. Through our teardown, we were able to infer the reasoning behind the placement of assets on the screen, the information hierarchy, and grouping principles, allowing us to build an idea of the concepts behind the design of Maps. The teardown also allowed us to analyze the visual design closely too, noting the rounded corners, captioning, bottom sheets, primary and secondary buttons, that Maps uses. We obtained many of the Maps’ assets through Figma community creators. For the notification cards, we obtained the animated images from Flaticon, a free-to-use icon database.

6.2 Onboarding

First, a user would be taken through an onboarding flow, which would provide instructions to the user on how to use the new feature, the permissions required for the feature, and how and where to find the feature. This helps the user get introduced to the new feature and learn the ways in which it monitors the environment.

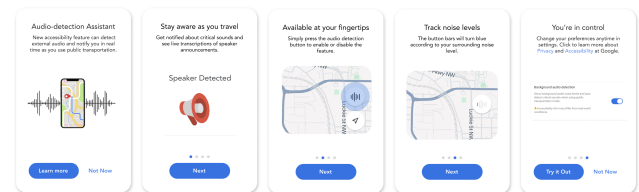


Figure 1: Onboarding flow depicting the newly added feature that user sees once.

6.3 Notification cards

These notification cards would appear on the screen when the system detects external audio from sources including a train, a bus, an emergency vehicle, a fire alarm, and a speaker. The system would use an AI model to detect and differentiate between the audio sources. The images are animated to visually communicate the urgency of the given notifying scenario. Additionally, for a speaker scenario, the system may transcribe any spoken audio it detects.

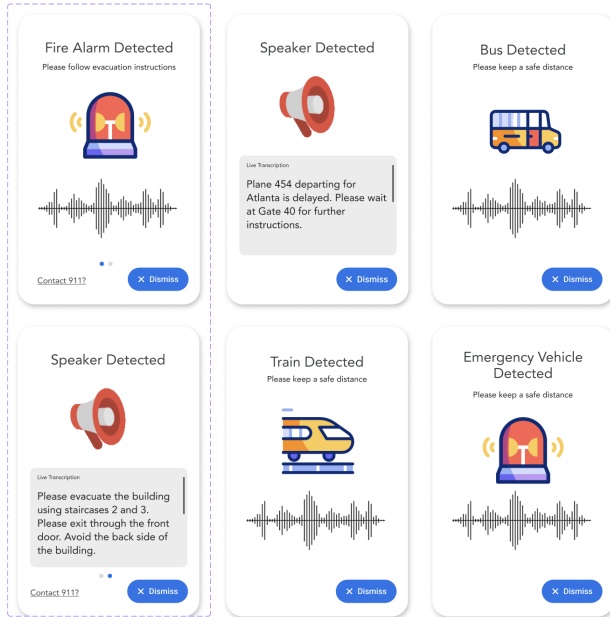


Figure 2: Notification alert cards that user sees when detected audio cues nearby

6.4 Audio History

The Audio History feature allows a user to see which audio cues have passed. Users will be able to see if they missed something, perhaps when their phone was kept away, when their journey was not active, or when they simply did not notice the notification.

6.5 User Audio Feedback

User Audio Feedback is a feature designed for a hearing individual. This flow goes through the validation process of alarming sounds like Fire Alarms, Smoke Alarms, or other Alarms, since human validation of sounds can help assure the DHH user. Since we cannot guarantee the accuracy of the AI model at such an early stage of the project, this is a means to validate audio cues, helping the feature to be more accurate.

7 Evaluation and Analysis

During our evaluation phase, we conducted sessions with five total experts — four of whom were MS-HCI students at Georgia Tech and one of whom was a DHH education specialist from Columbia

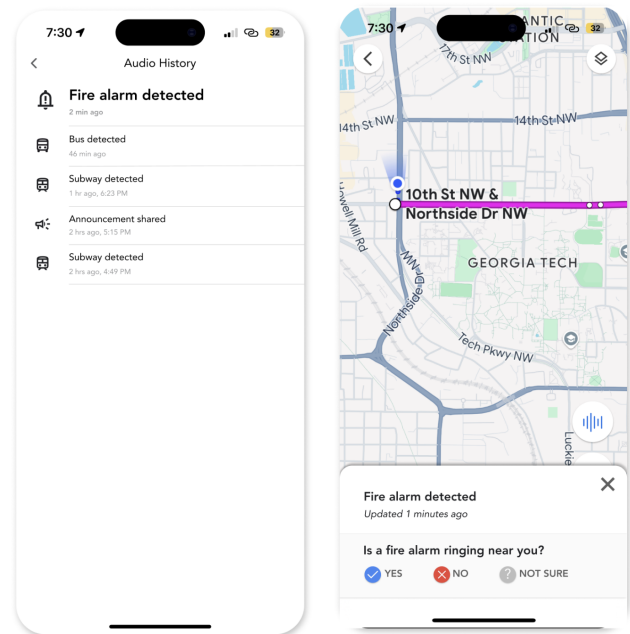


Figure 3: (Left) Audio History for a Navigation; (Right) User Audio Feedback Popup

University. We believed that the HCI students could provide valuable feedback on the overall design and that the education specialist could offer perspective into the specific needs of DHH users. Our sessions consisted of a cognitive walkthrough and a heuristic evaluation.

7.1 Analysis of Cognitive Walkthroughs

During the cognitive walkthrough, we instructed the evaluators to complete a series of tasks with our prototype. Our data consisted of the real-time comments and feedback our evaluators gave as they went through each task. Overall, our participants were impressed by the prototype, noting how well it “integrates with Google’s current style.” However, we also found that our prototype had small UI inconsistencies that created friction in the design, for example, a participant noted that “the spacing for ‘Contact 911’ is off.” Many of the experts felt that color was not the best indicator for information with a participant suggesting it was “the worst indicator for UI stuff” – this was especially problematic for the sound level feature, by which sound level was initially represented by red (a lot of surrounding noise), yellow (moderate surrounding noise), and green (minimal surrounding noise).

7.2 Analysis of Heuristic Evaluation

After the cognitive walkthrough, we had the participant fill out a heuristic evaluation form, where they evaluated the prototype based on a set of criteria and gave scores from 0 to 5 (the highest possible score) for specific sub-criteria. The criteria were determined based on Nielsen’s 10 design Heuristics [19]. For analysis we first took averages of sub-criteria for each heuristic from each evaluator and then averaged these averages to obtain average scores for each

heuristic. We obtained an average score of 4.225, indicating an overall effective experience.

7.3 User Group Testing

Initially, we intended to conduct in-person interviews with the DHH community at the Georgia Center for Deaf and Hard of Hearing, but due to budget constraints for an interpreter, we created a captioned video demo and survey for feedback; however, we only received 3 responses. The survey asked participants to watch a video demo and rate the prototype on a 1-5 scale based on specific criteria, yielding an average score of 3.39. However, no participants provided comments or expressed interest in an interview, leaving the results unclear. Additionally, the survey format may not have been fully accessible to individuals with varying comfort levels or fluency in reading captions.

8 Conclusion

Our team has conceptualized a feature that may support DHH individuals as they navigate auditorily complex public transportation systems. After conducting in-depth interviews with DHH identifying individuals, leaders within DHH education, and accessibility experts, we discovered the many pain points and unmet needs for DHH individuals in these spaces. We used insights from these interviews to inform our design requirements, from which we drafted, prototyped, and iterated upon multiple versions of the audio detection feature. We decided to build and situate our system on Google Maps, leveraging the platform's accessibility and widespread use (particularly among the DHH community). To ensure that our system was meeting the needs of our intended user group, we worked alongside members of the DHH community and participants with expertise in design or DHH accessibility throughout our entire design process.

Our next goal is to develop and implement this design into an addition onto Google Maps through integrating the AI model. We began our focus in Atlanta, GA, but we see this system as useful to many cities that rely on public transportation, especially New York, San Francisco, and Chicago.

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